

Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Given

A generating circle of diameter 30 mm rolls internally without slipping inside a directing circle of diameter 150 mm.

- Construct one complete hypocycloid arch.
- Divide the arch into 12 equal angular parts.
- Retain every construction element in all later views.

Calculated Data

$$R = \frac{150}{2} = 75 \text{ mm},$$

$$OC_0 = R - r = 75 - 15 = 60 \text{ mm},$$

$$\angle AOB = \frac{360^\circ r}{R} = \frac{360^\circ(15)}{75} = 72^\circ,$$

$$\Delta\theta = \frac{72^\circ}{12} = 6^\circ,$$

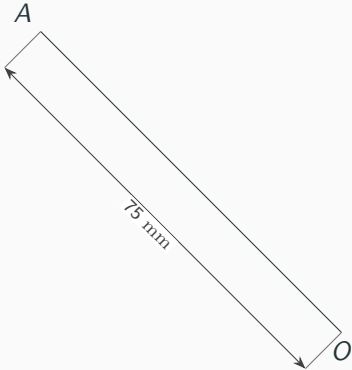
$$r = \frac{30}{2} = 15 \text{ mm},$$

$$AC_0 = 15 \text{ mm},$$

$$\text{Generating-circle division} = \frac{360^\circ}{12} = 30^\circ.$$

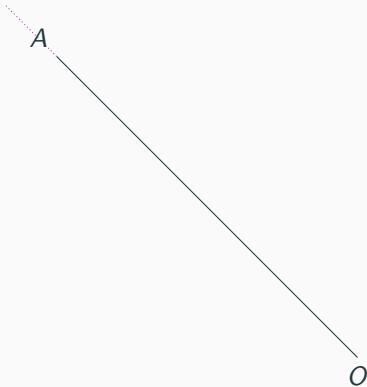
Step 1: Draw OA

$OA = 75 \text{ mm.}$



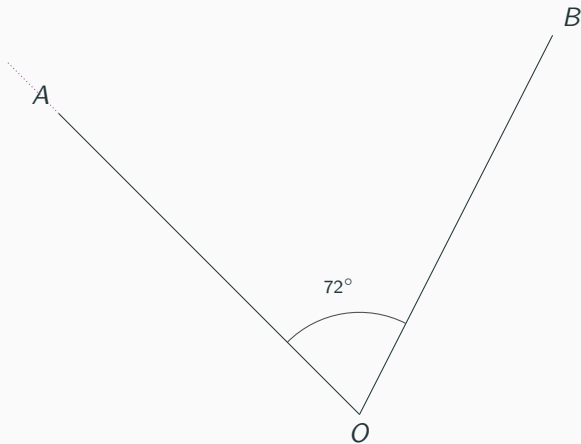
Step 2: Extend OA

Extend OA .

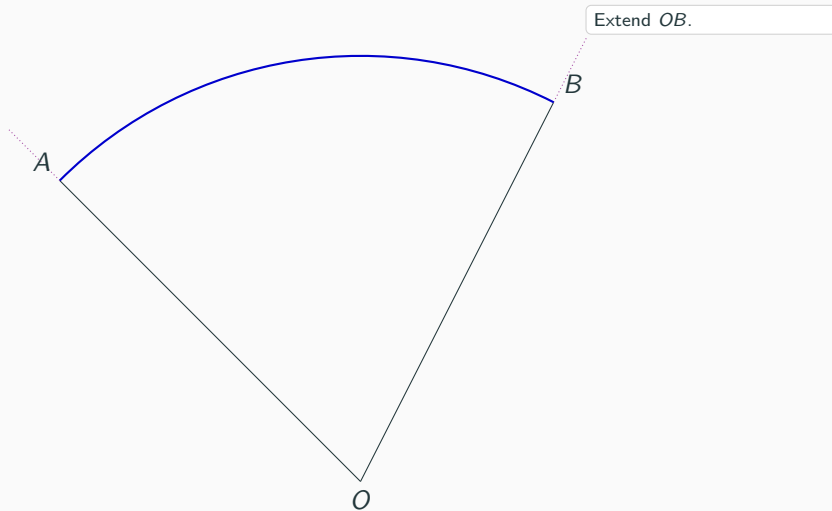


Step 3: Draw OB at 72°

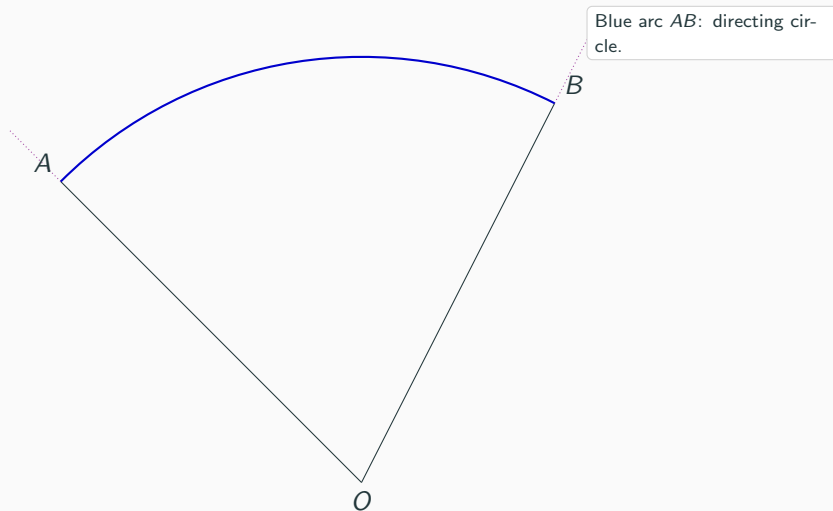
$$\angle AOB = 72^\circ.$$



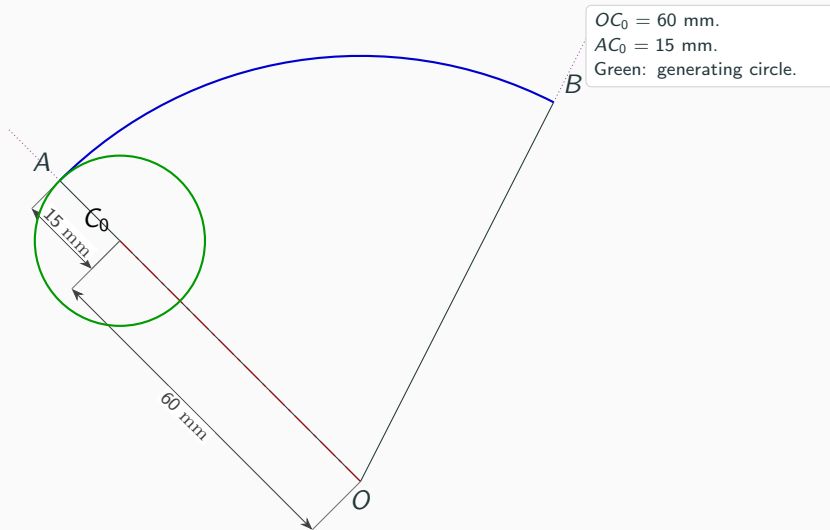
Step 4: Extend OB



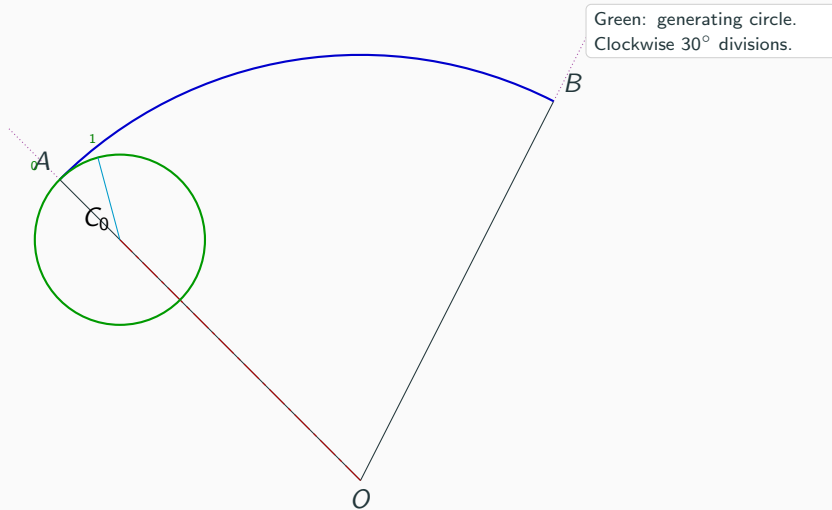
Step 5: Draw the directing arc AB



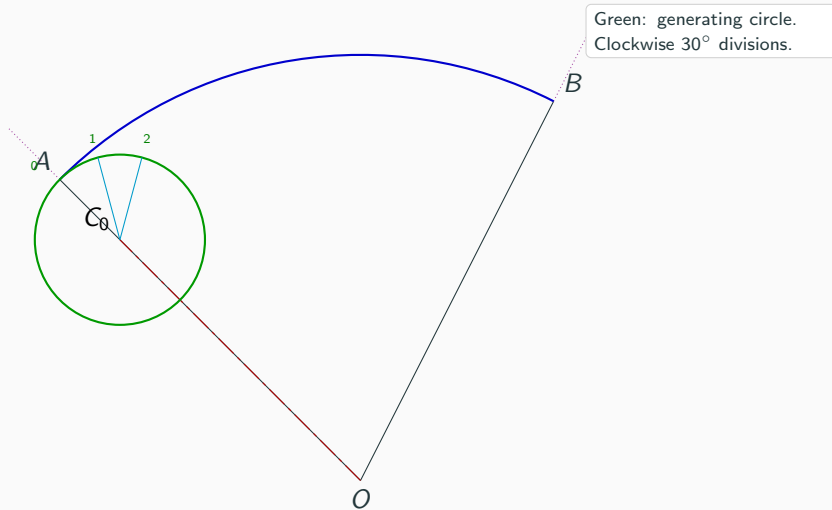
Step 6: Locate C_0 and the generating circle



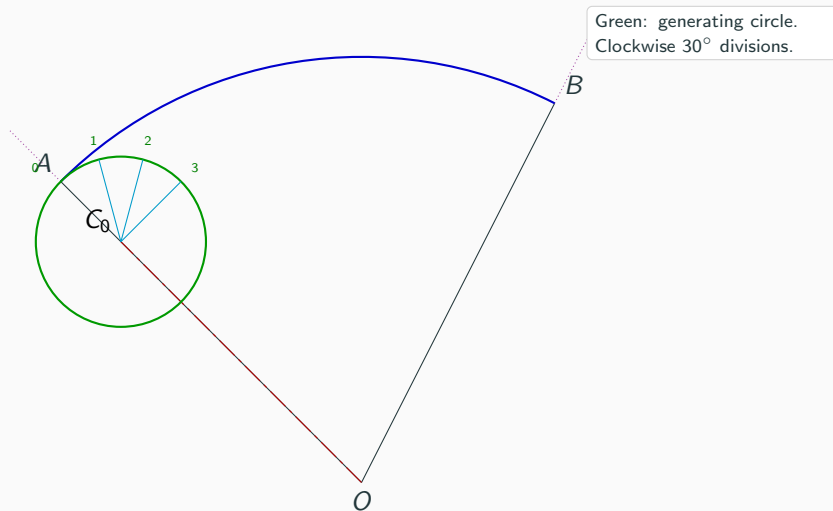
Step 7: Divide the generating circle into 12 equal parts



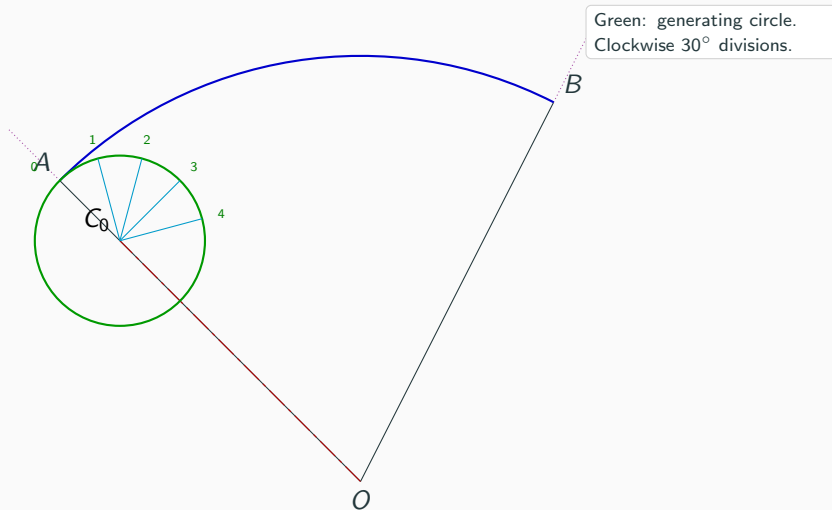
Step 7: Divide the generating circle into 12 equal parts



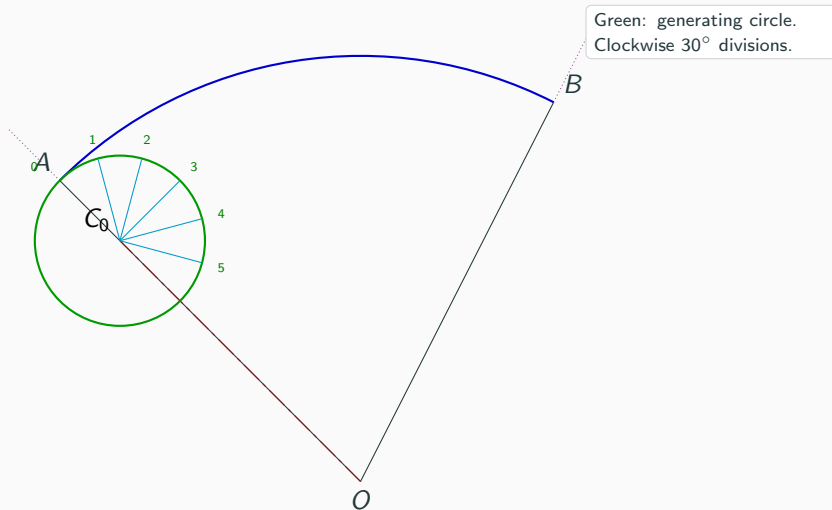
Step 7: Divide the generating circle into 12 equal parts



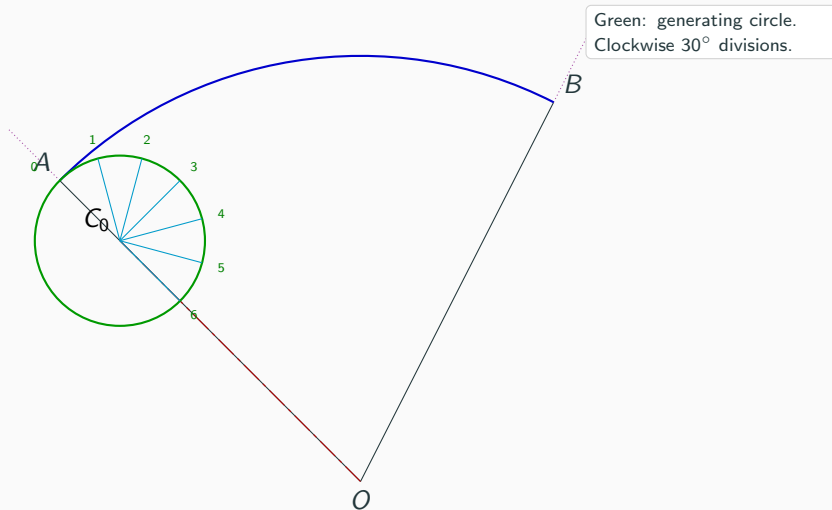
Step 7: Divide the generating circle into 12 equal parts



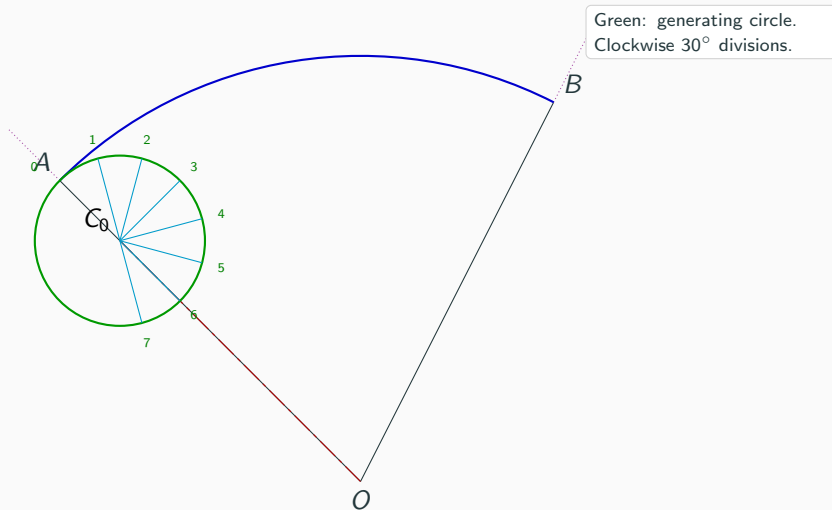
Step 7: Divide the generating circle into 12 equal parts



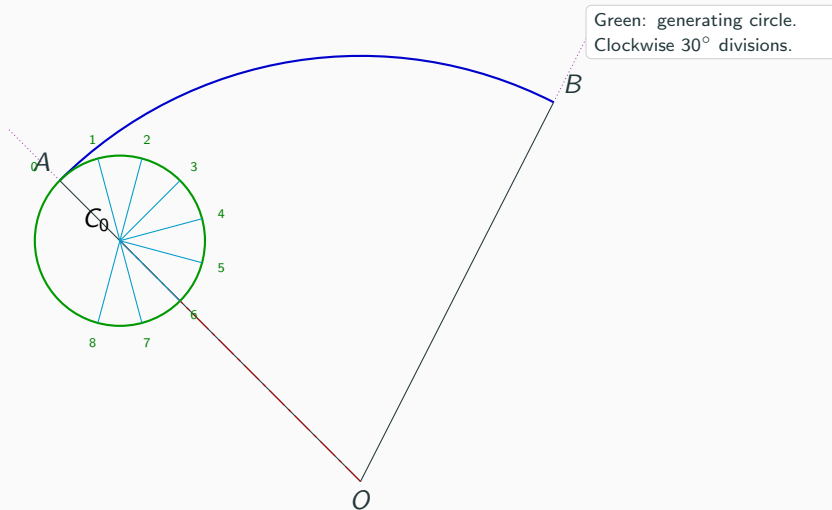
Step 7: Divide the generating circle into 12 equal parts



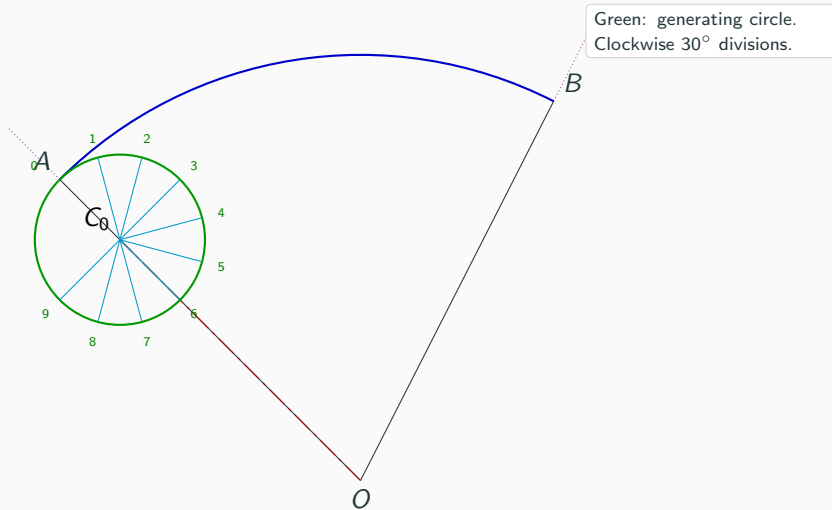
Step 7: Divide the generating circle into 12 equal parts



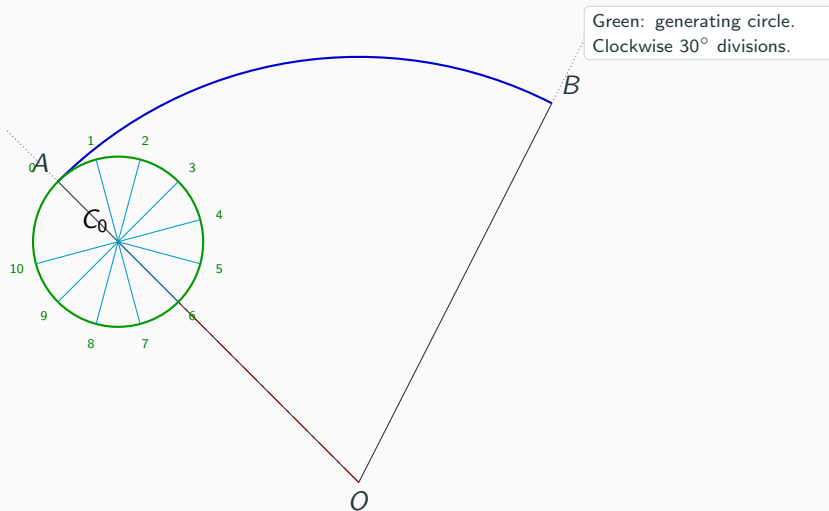
Step 7: Divide the generating circle into 12 equal parts



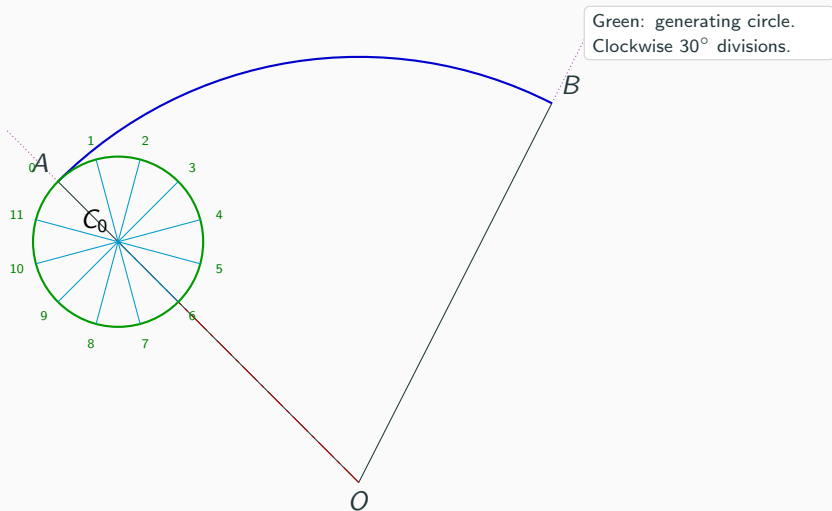
Step 7: Divide the generating circle into 12 equal parts



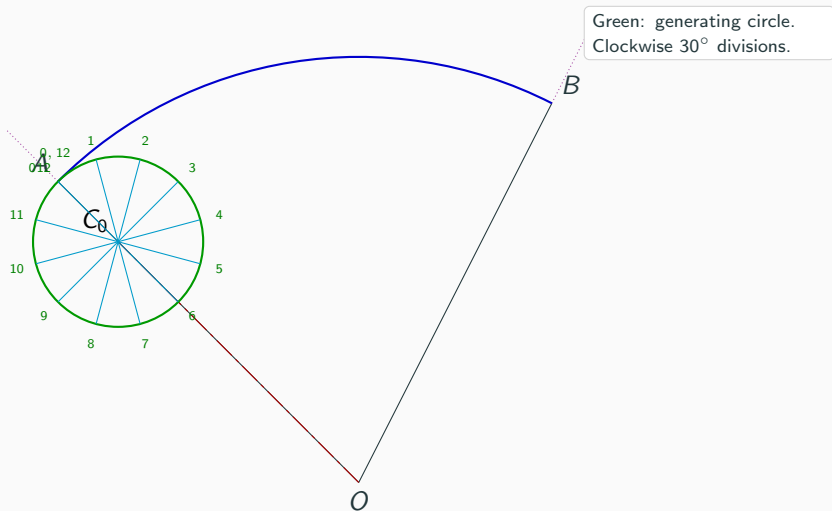
Step 7: Divide the generating circle into 12 equal parts



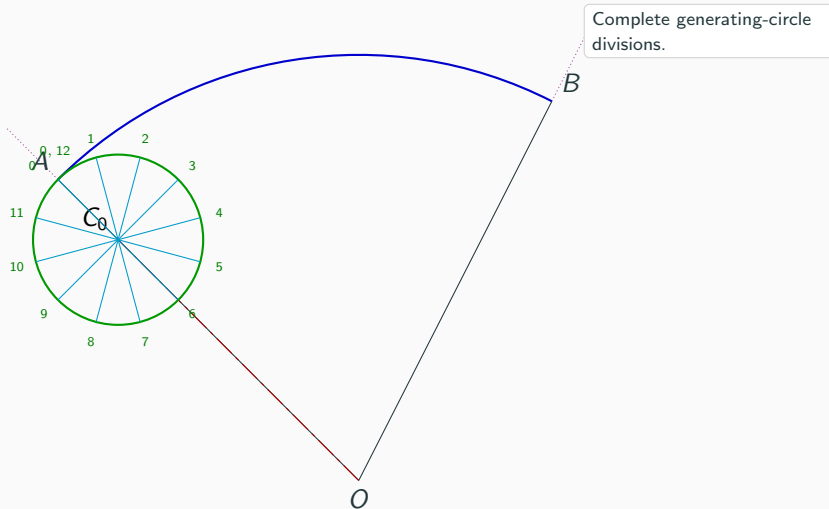
Step 7: Divide the generating circle into 12 equal parts



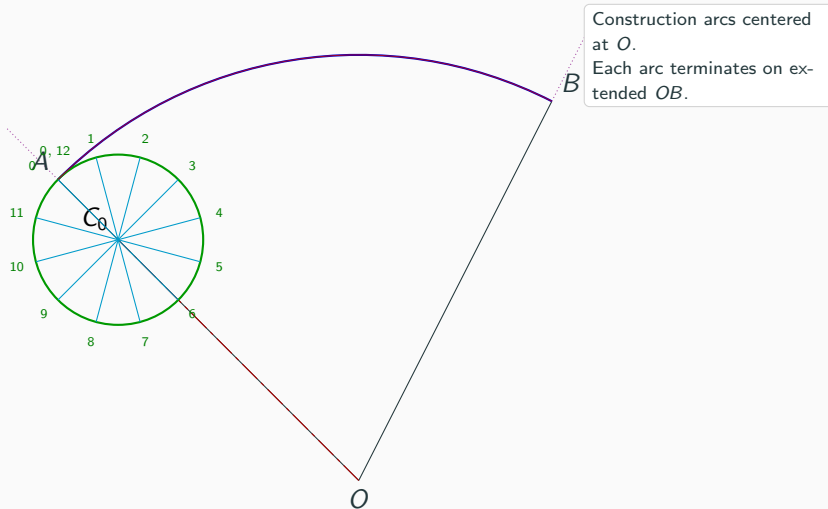
Step 7: Divide the generating circle into 12 equal parts



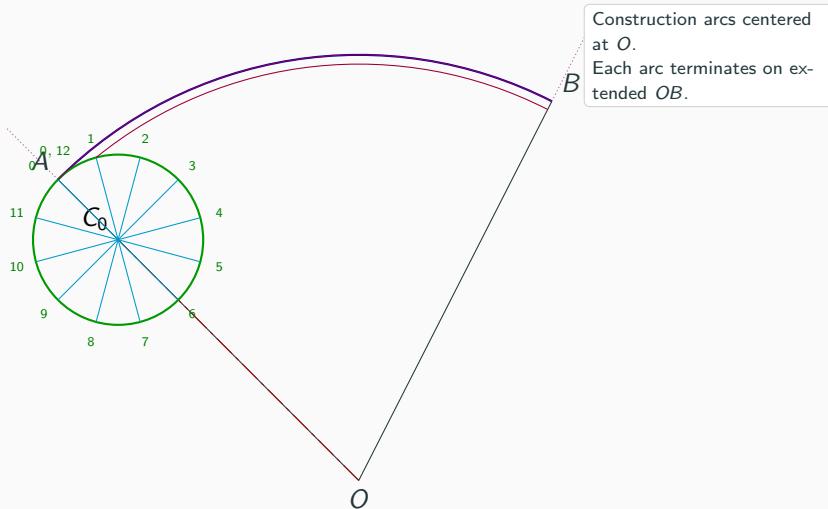
Step 8: Complete the generating-circle divisions



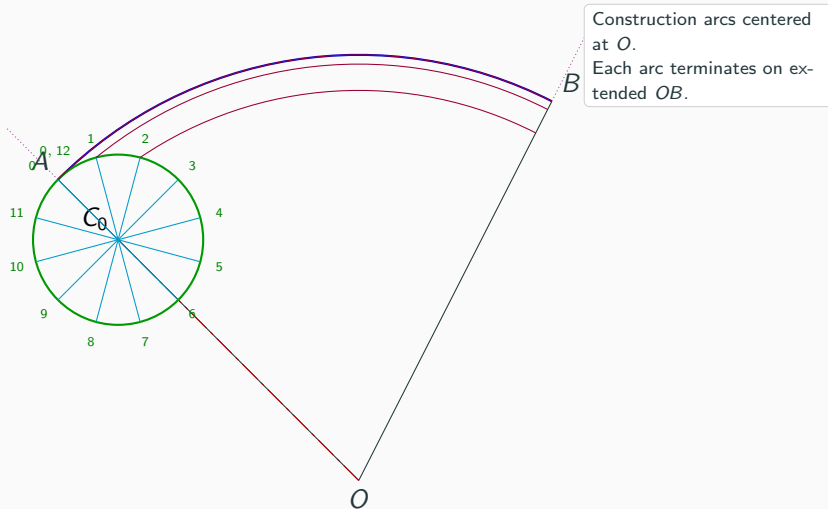
Step 9: Draw concentric construction arcs



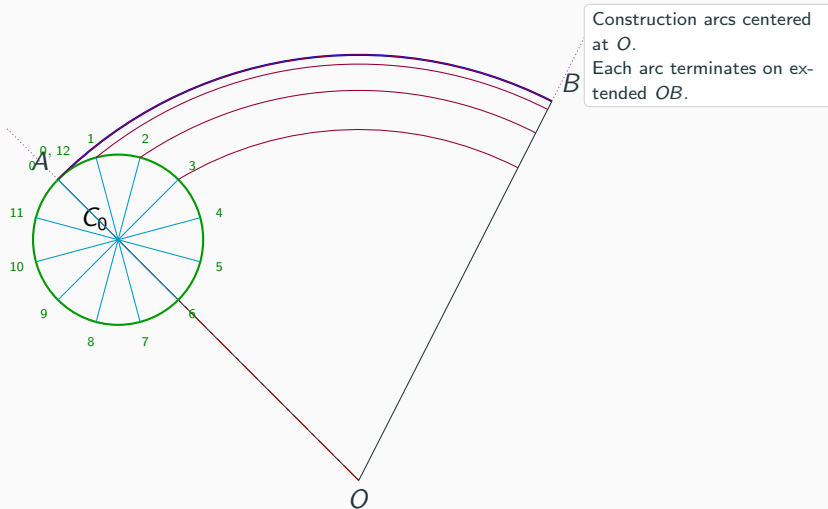
Step 9: Draw concentric construction arcs



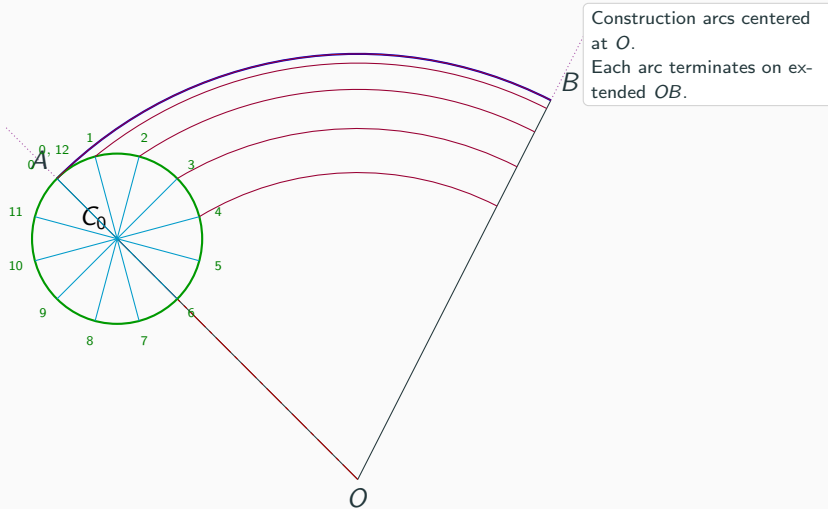
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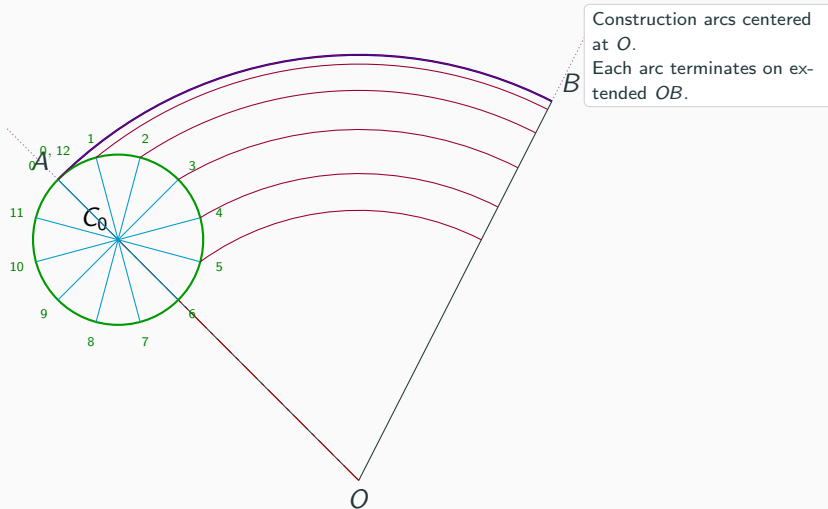
Step 9: Draw concentric construction arcs



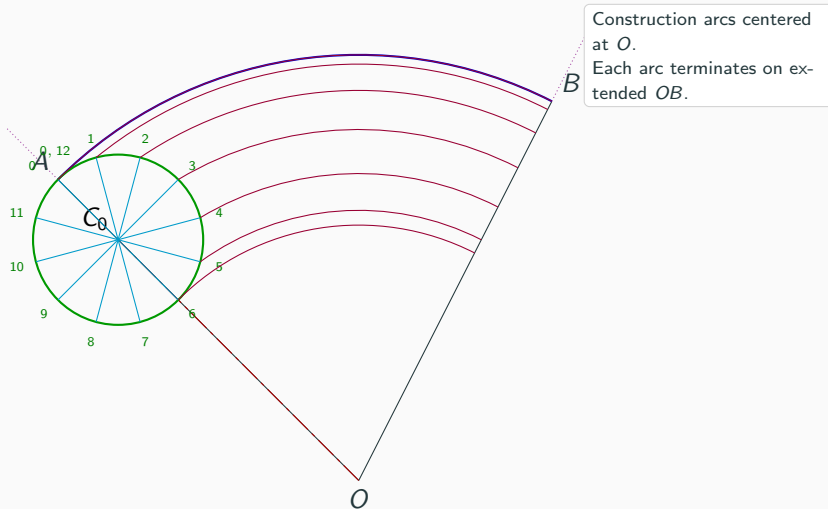
Step 9: Draw concentric construction arcs



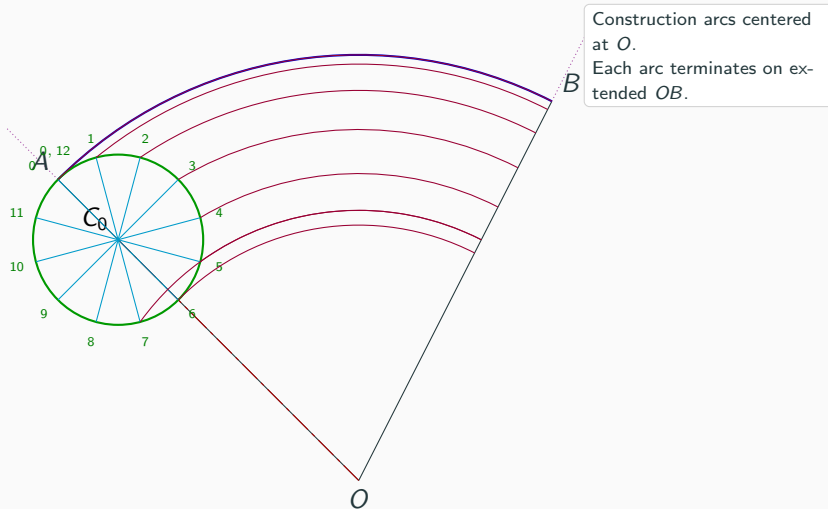
Step 9: Draw concentric construction arcs



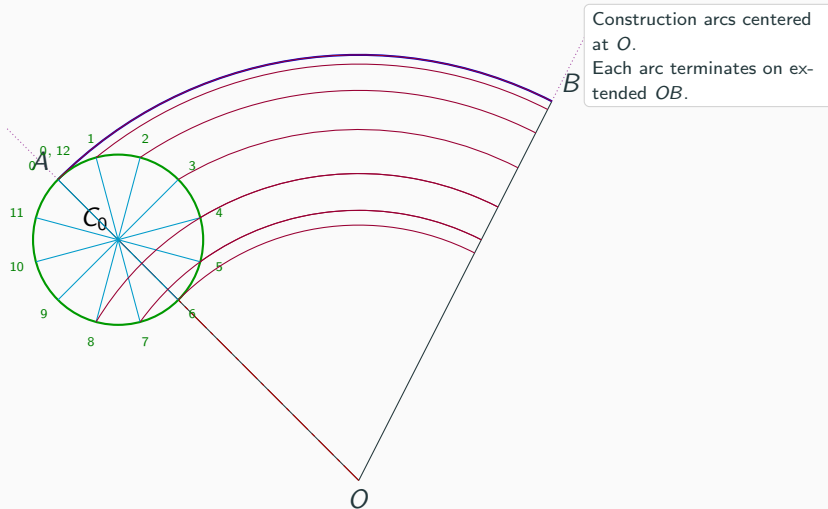
Step 9: Draw concentric construction arcs



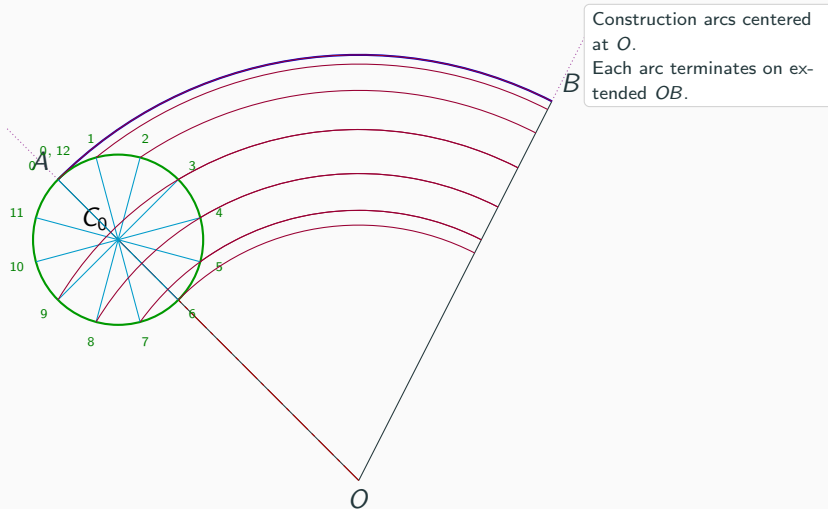
Step 9: Draw concentric construction arcs



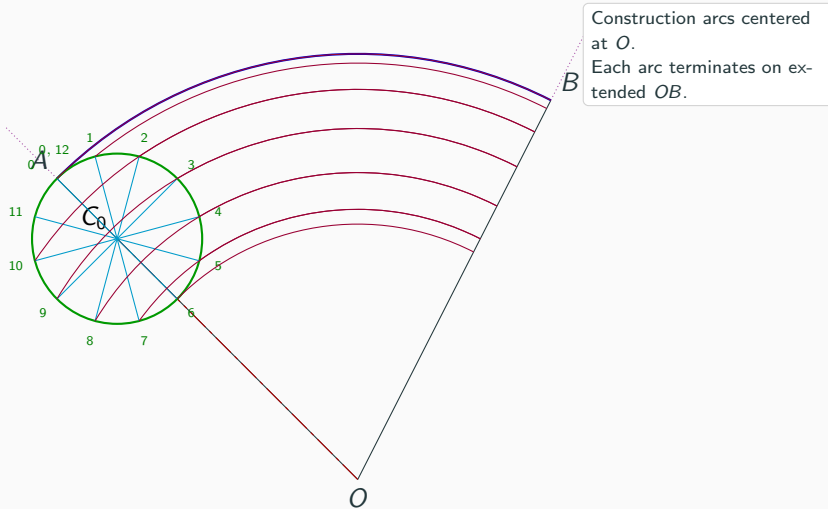
Step 9: Draw concentric construction arcs



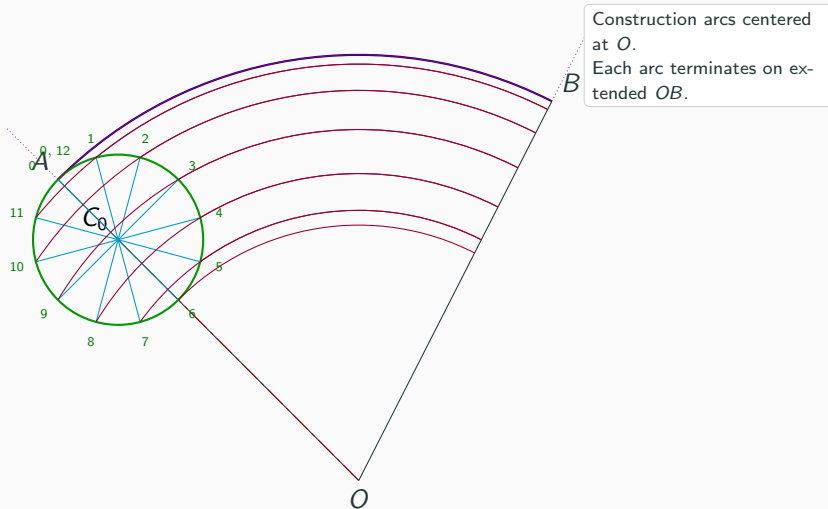
Step 9: Draw concentric construction arcs



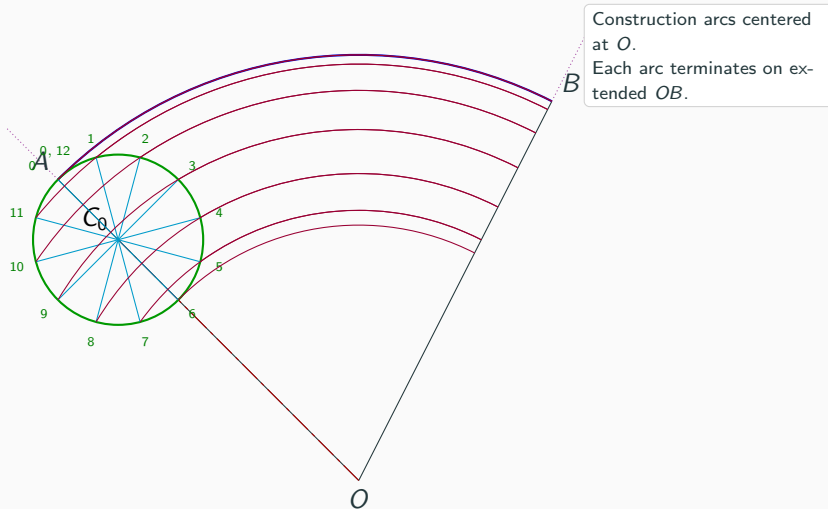
Step 9: Draw concentric construction arcs



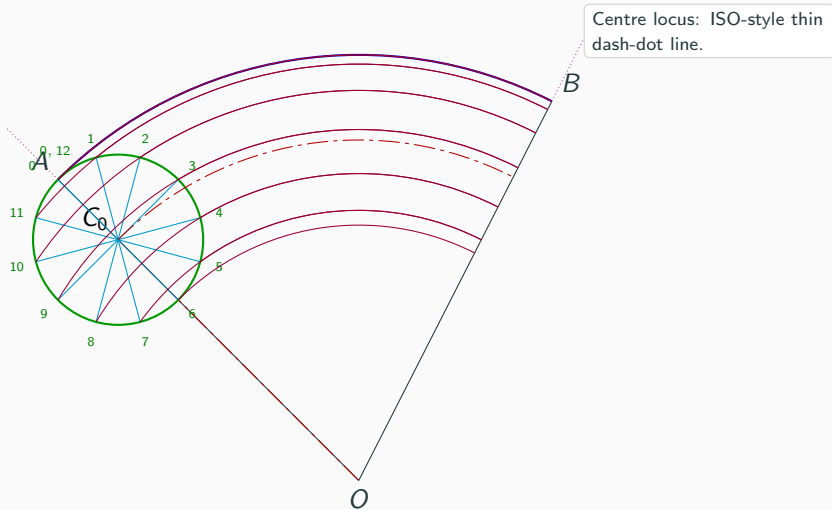
Step 9: Draw concentric construction arcs



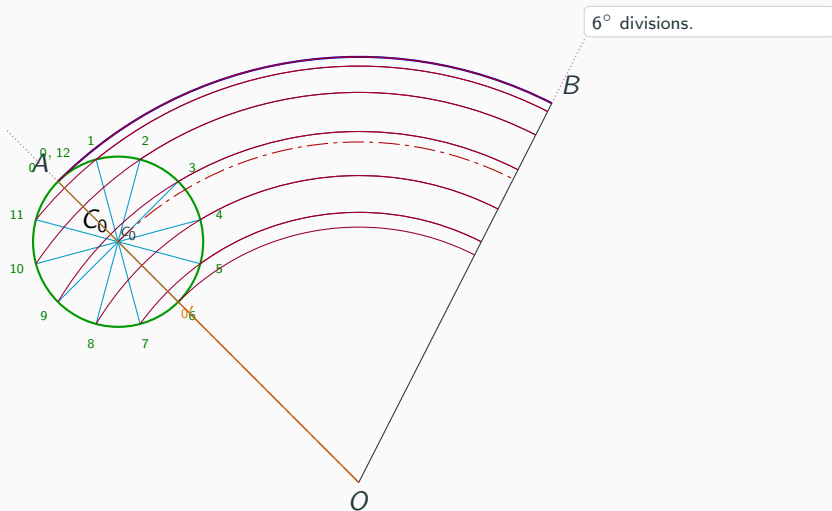
Step 9: Draw concentric construction arcs



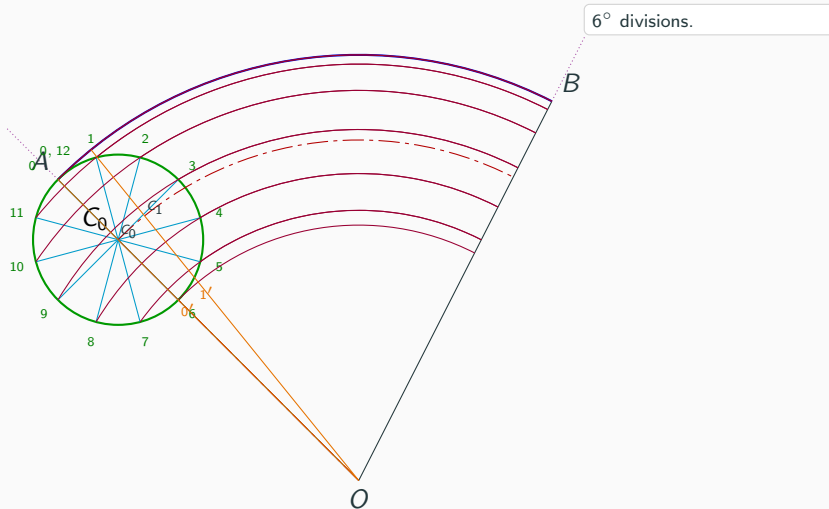
Step 10: Draw the centre arc through C_0



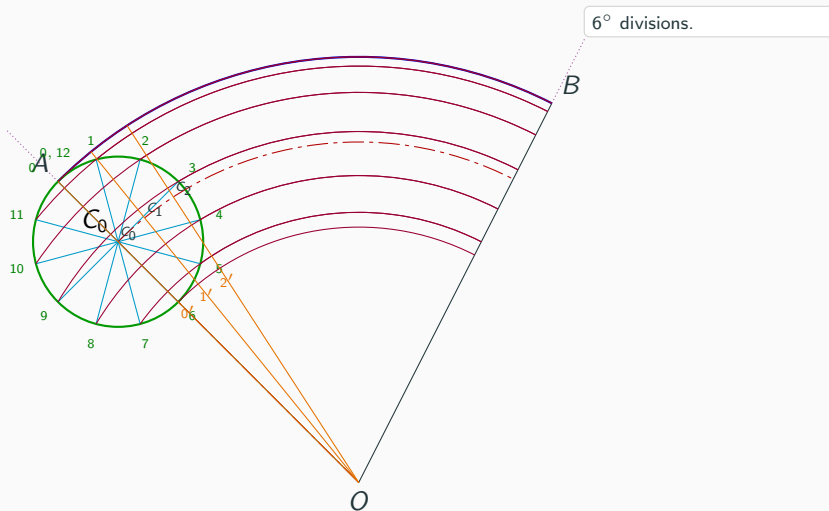
Step 11: Divide $\angle AOB$ into 12 equal parts



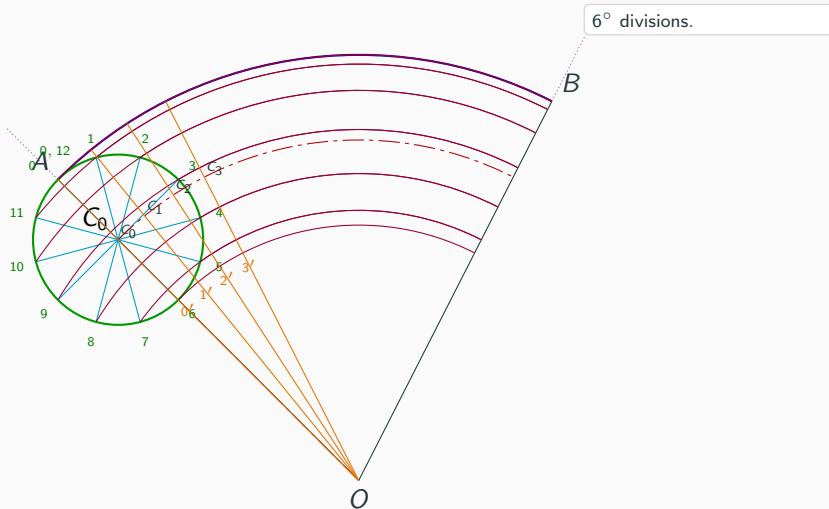
Step 11: Divide $\angle AOB$ into 12 equal parts



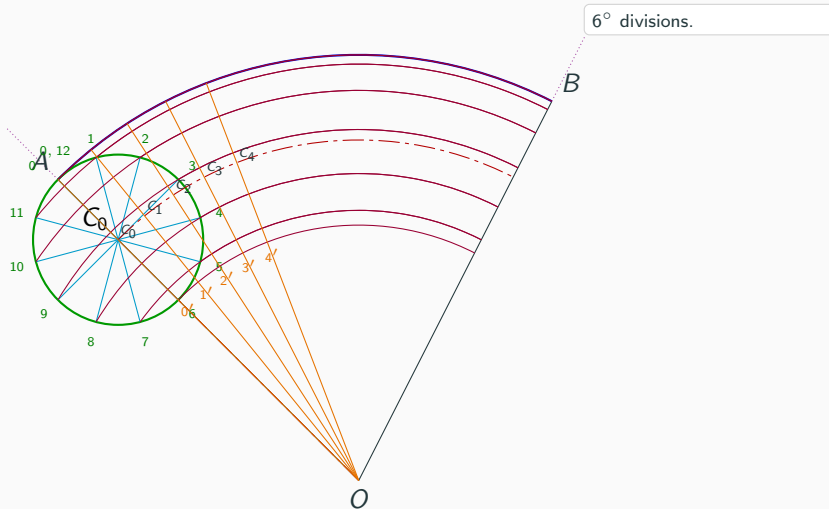
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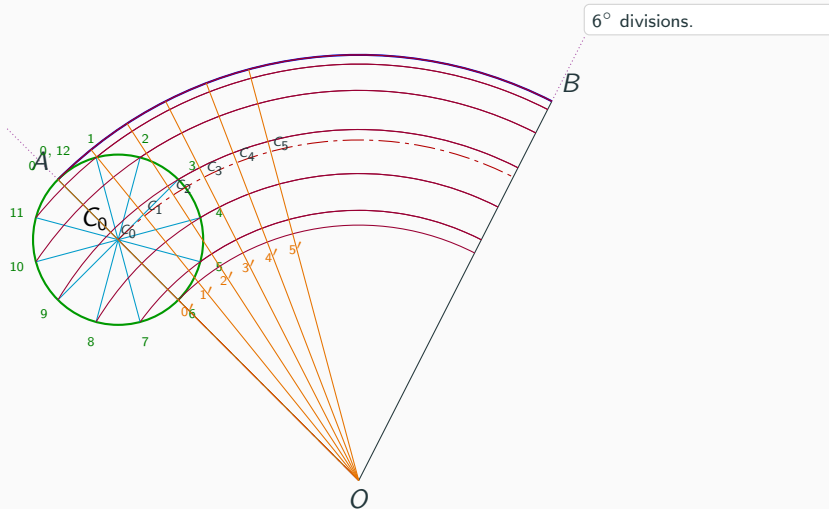
Step 11: Divide $\angle AOB$ into 12 equal parts



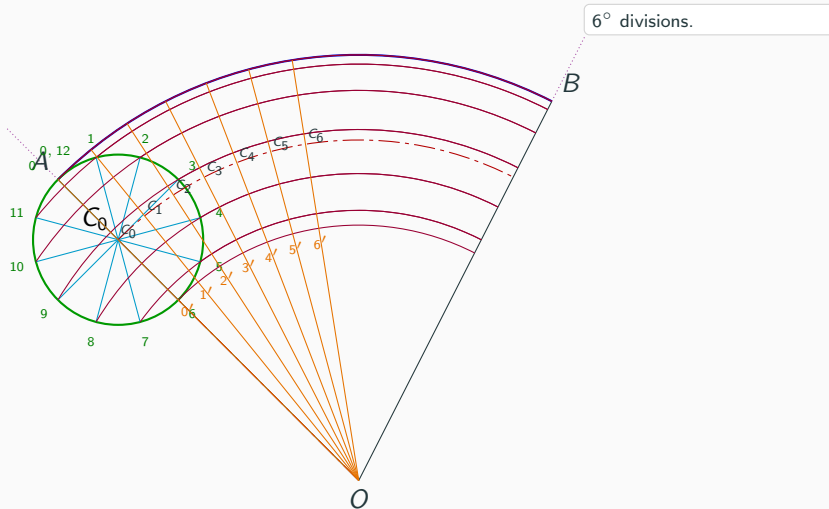
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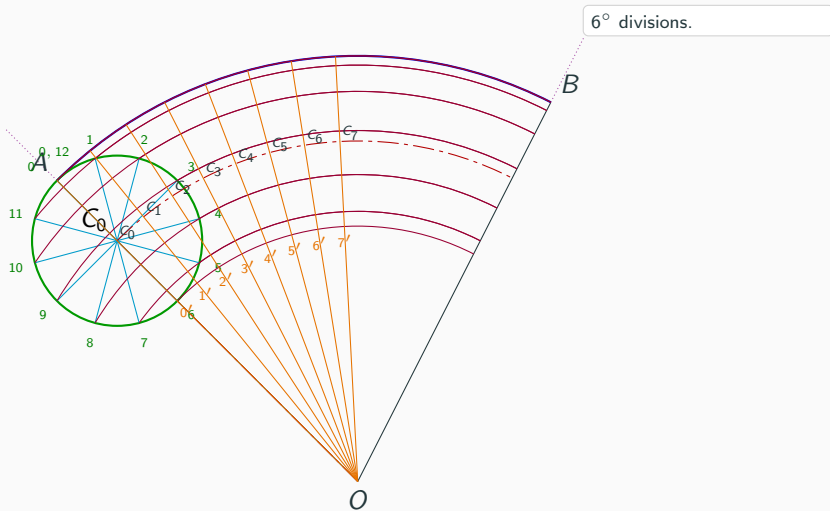
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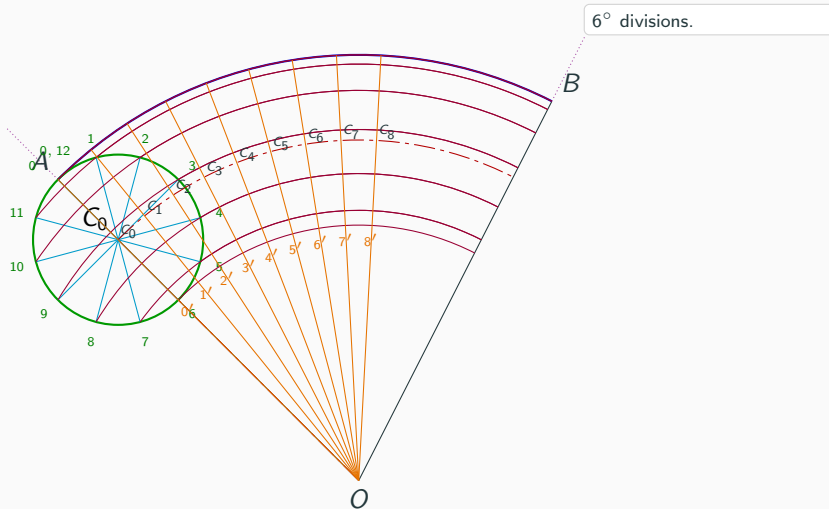
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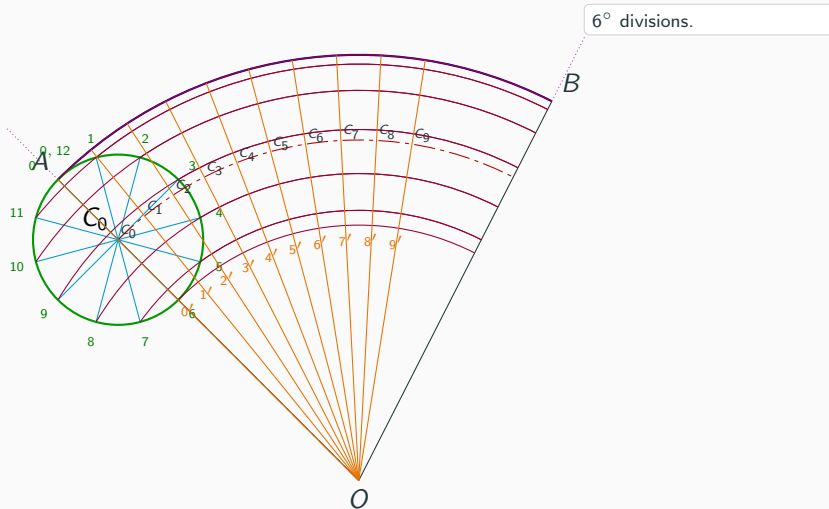
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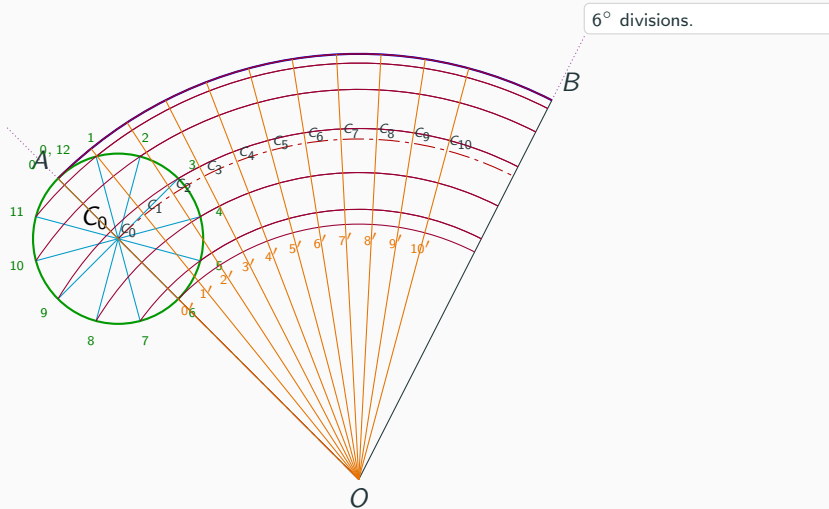
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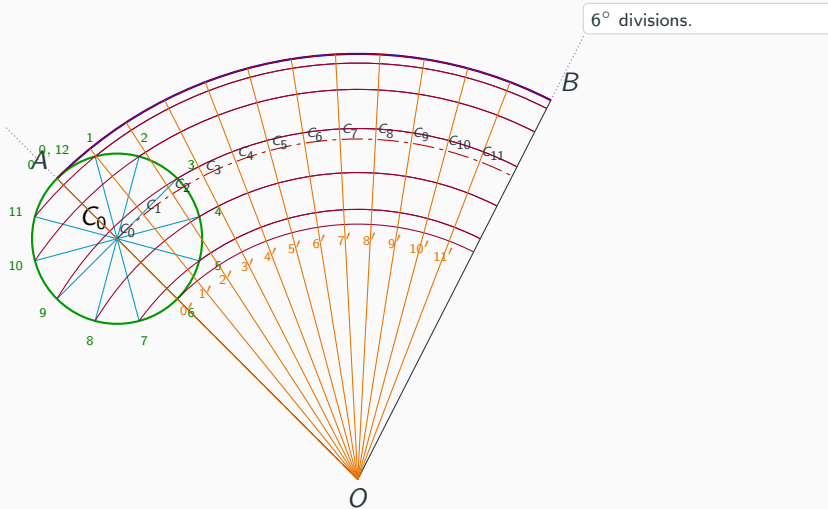
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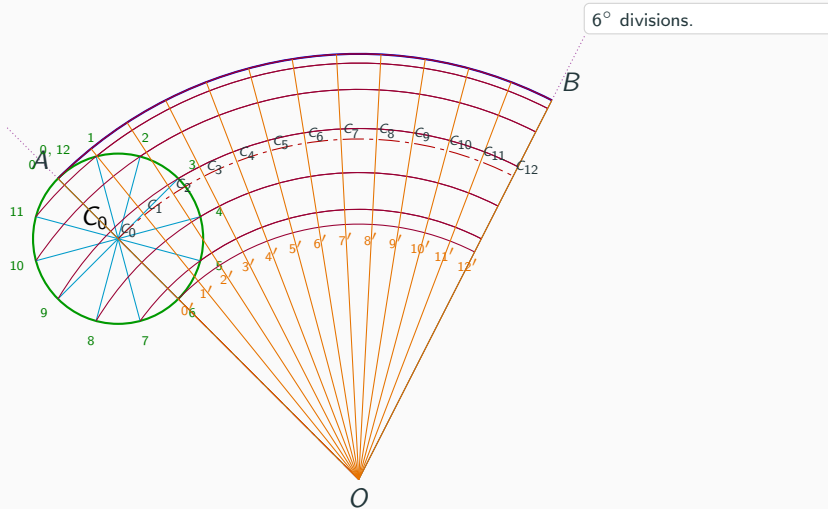
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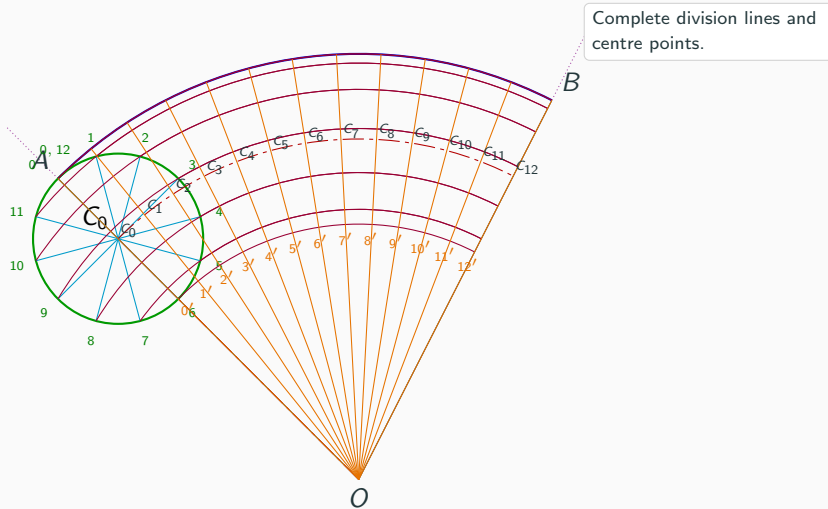
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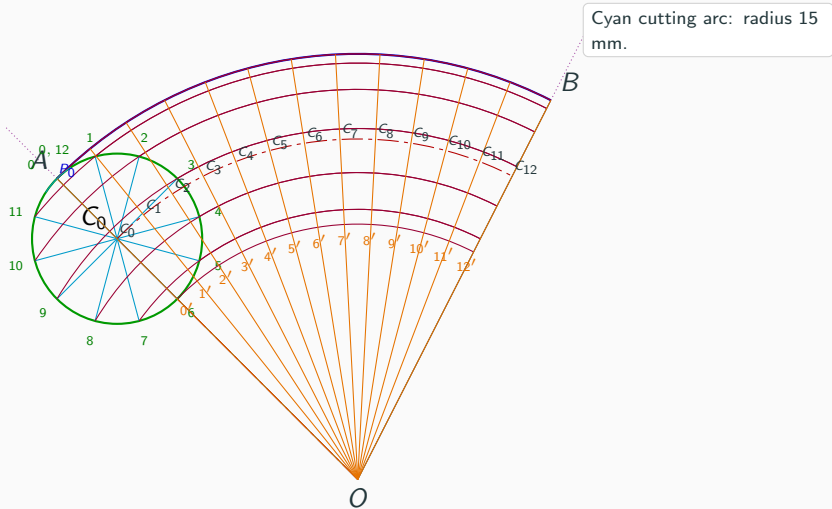
Step 11: Divide $\angle AOB$ into 12 equal parts



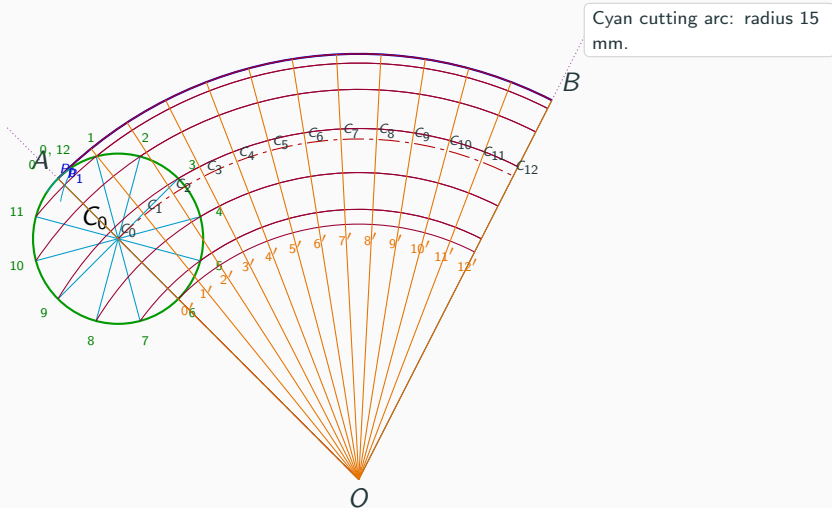
Step 12: Complete the angular divisions and centre points



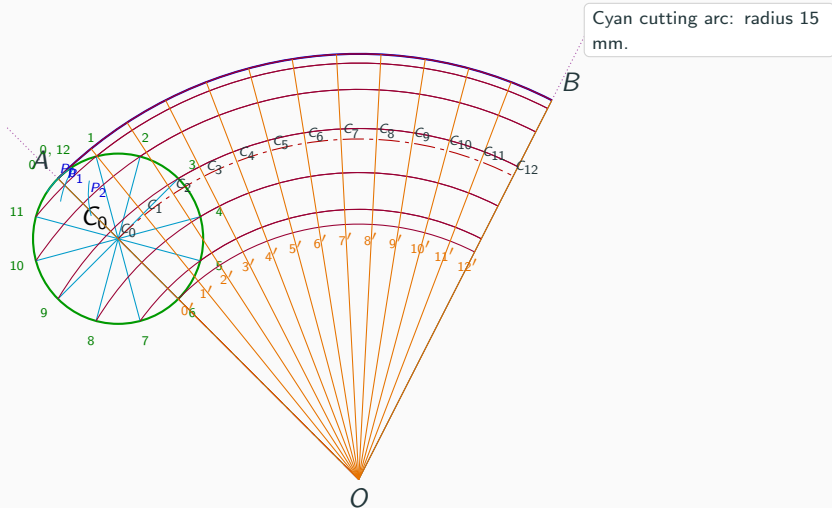
Step 13: Draw the cutting arc and locate P_i



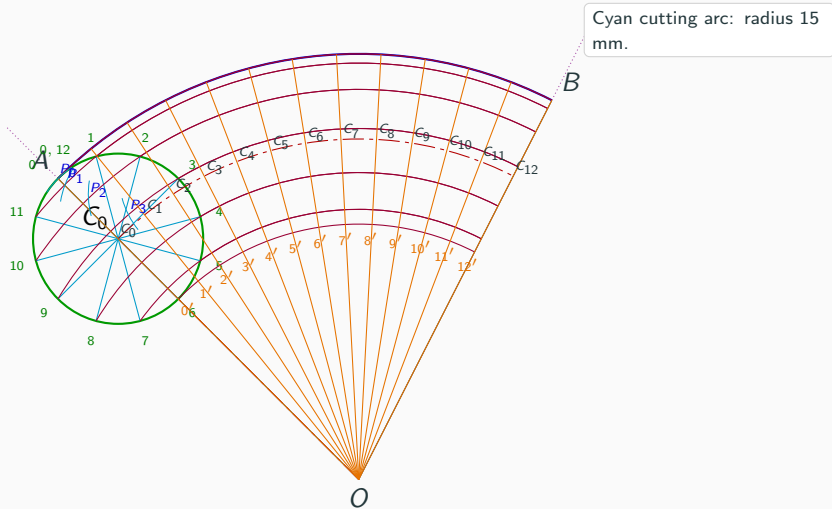
Step 13: Draw the cutting arc and locate P_i



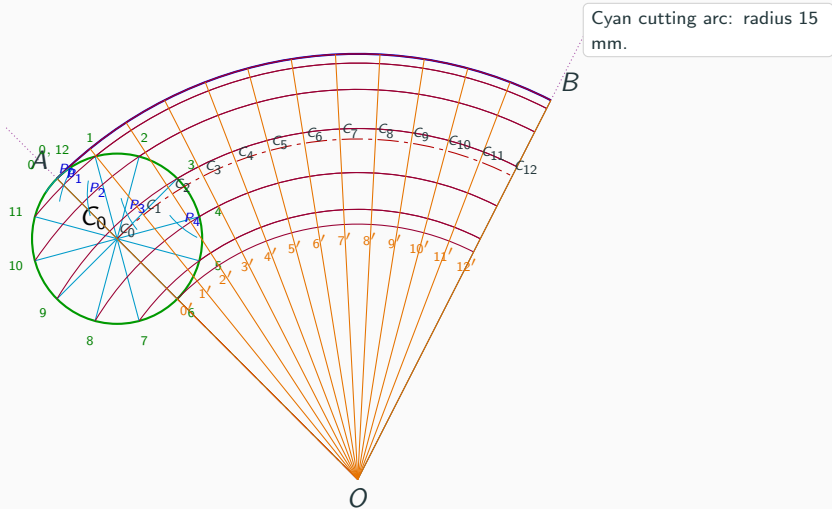
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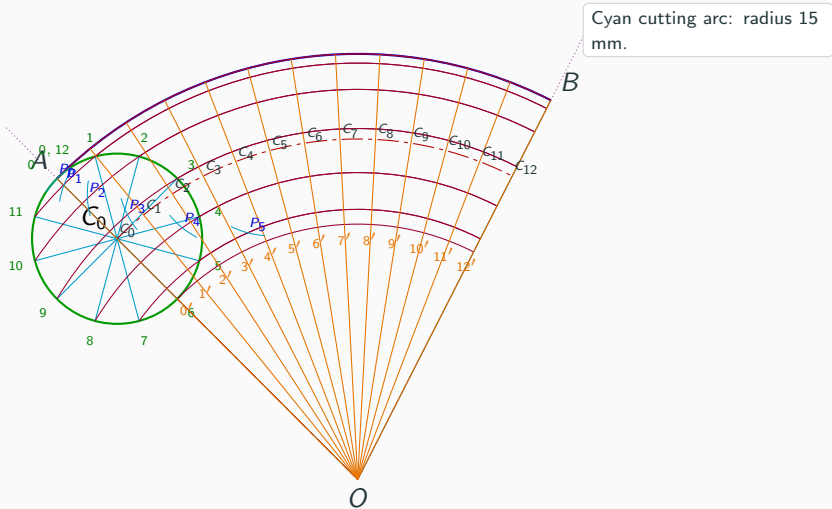
Step 13: Draw the cutting arc and locate P_i



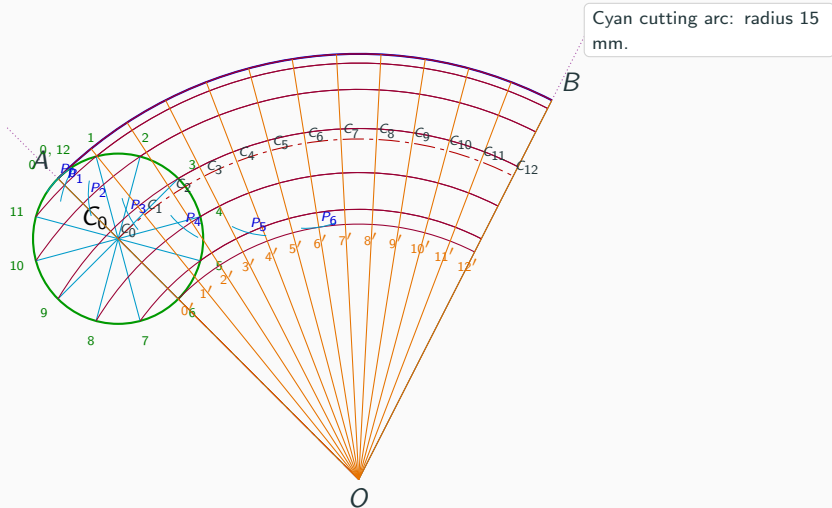
Step 13: Draw the cutting arc and locate P_i



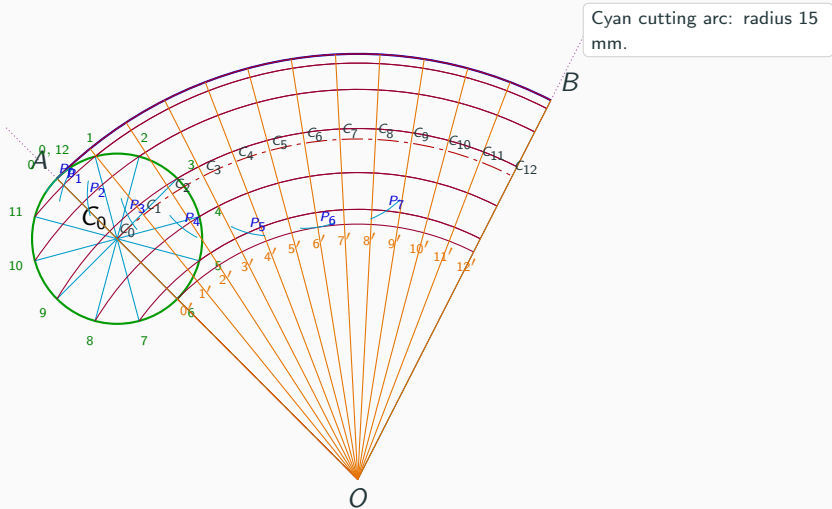
Step 13: Draw the cutting arc and locate P_i



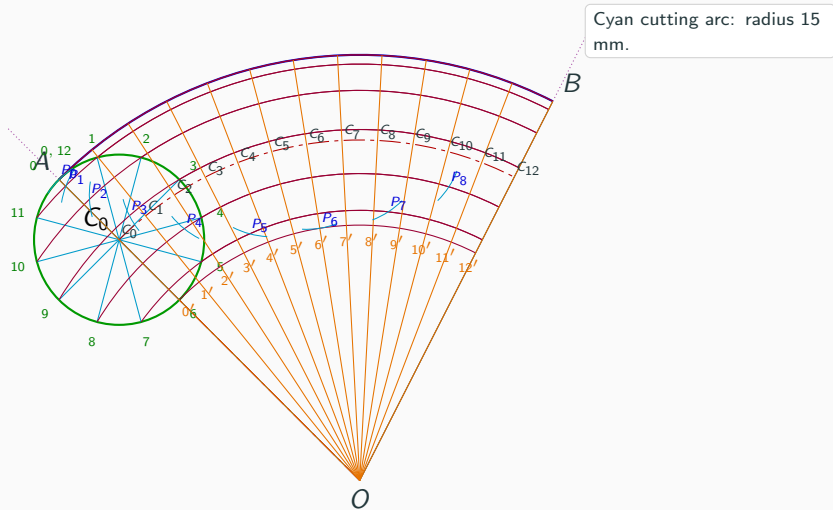
Step 13: Draw the cutting arc and locate P_i



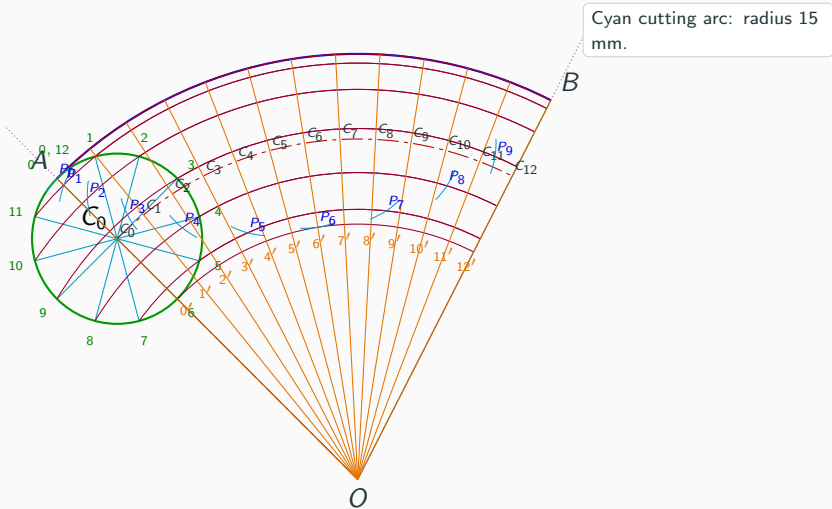
Step 13: Draw the cutting arc and locate P_i



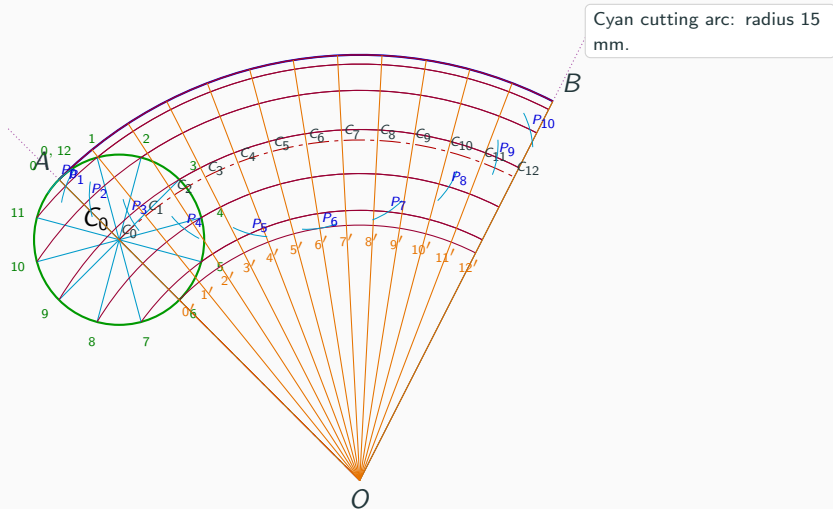
Step 13: Draw the cutting arc and locate P_i



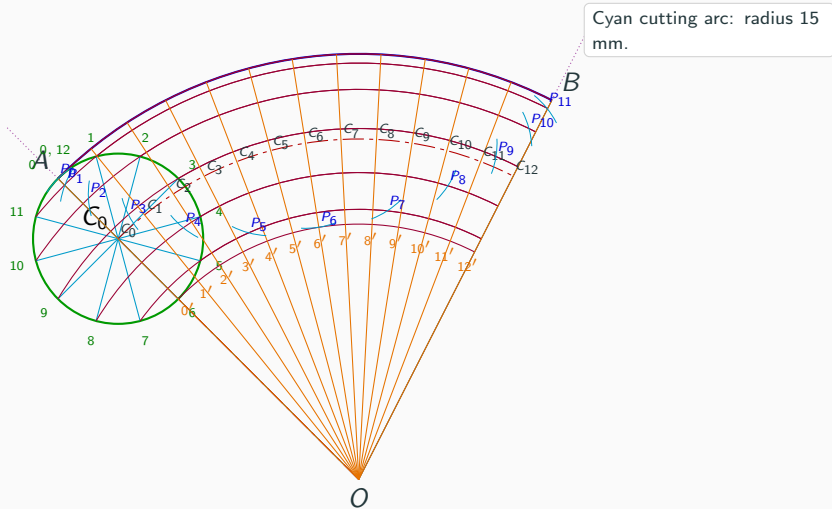
Step 13: Draw the cutting arc and locate P_i



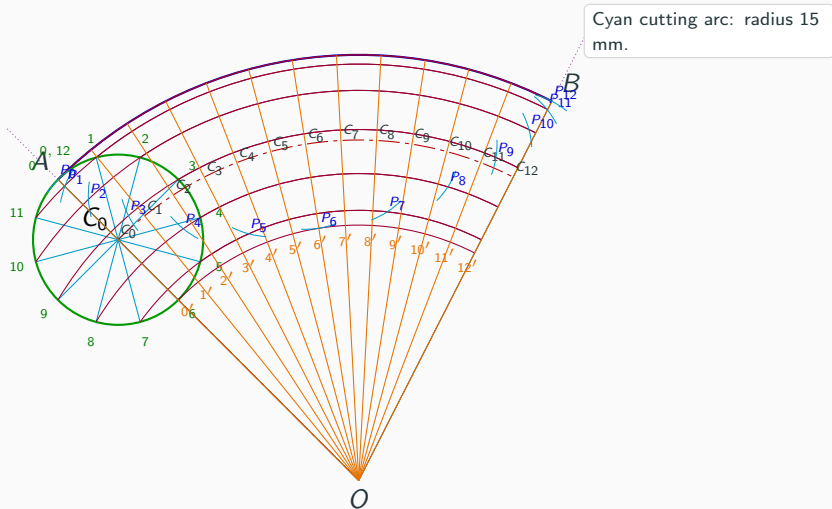
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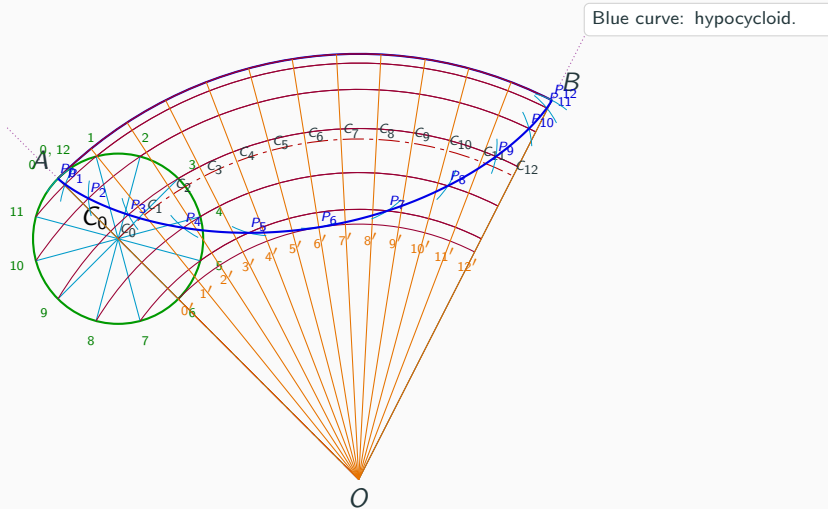
Step 13: Draw the cutting arc and locate P_i



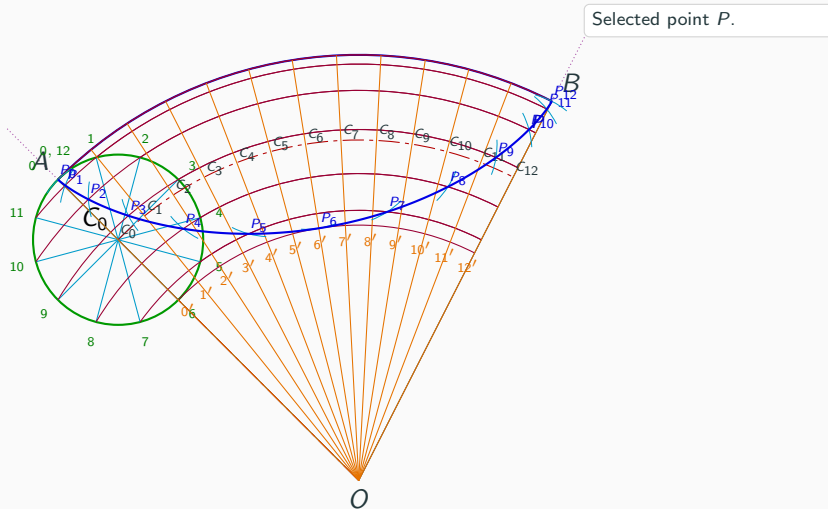
Step 13: Draw the cutting arc and locate P_i



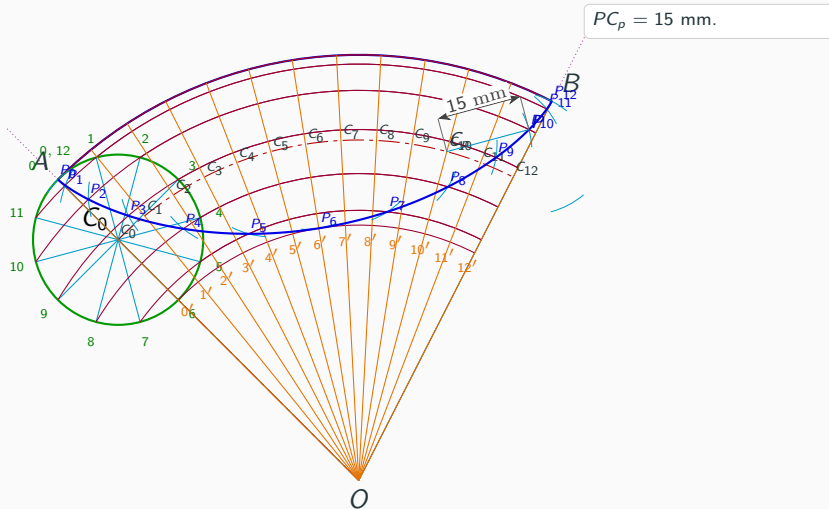
Step 14: Draw the hypocycloid



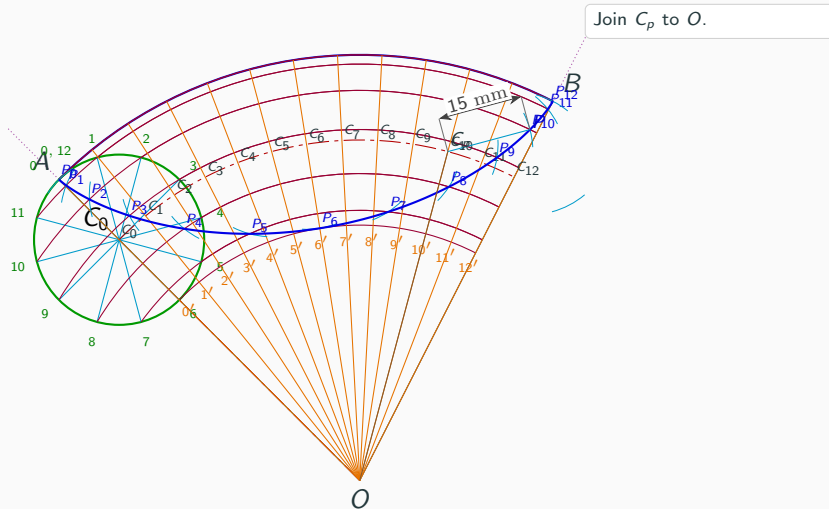
Step 15: Select a point P on the hypocycloid



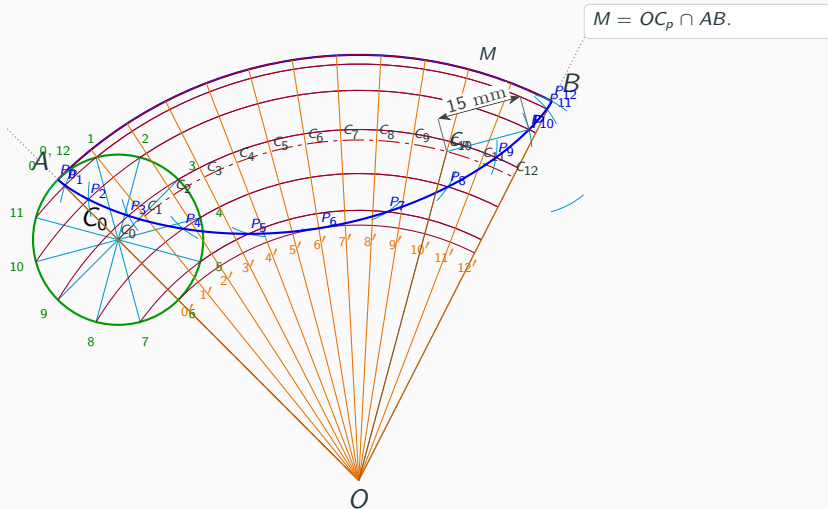
Step 16: Draw the 15 mm arc from P to the centre locus



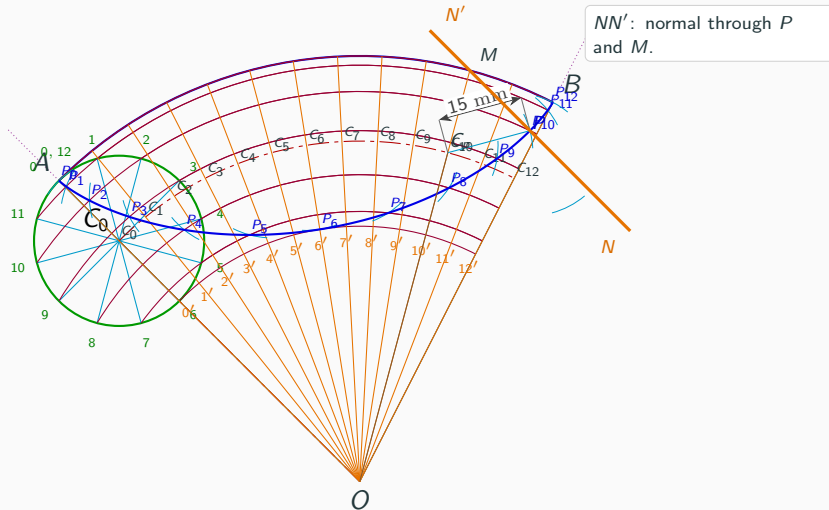
Step 17: Join C_p to O



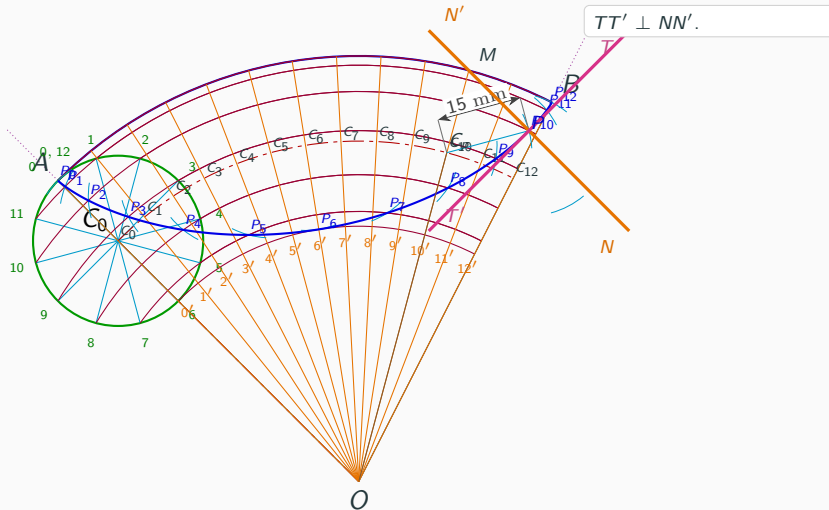
Step 18: Mark M on the directing circle



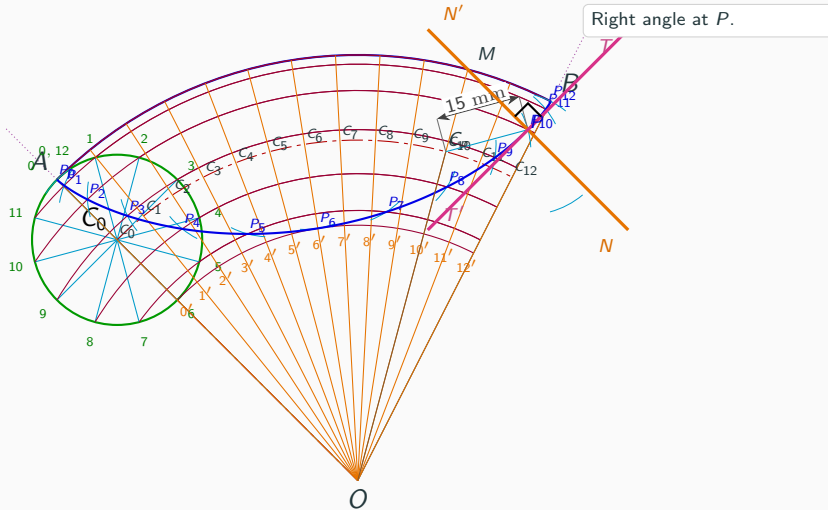
Step 19: Draw the normal NN' through P



Step 20: Draw the tangent TT' perpendicular to NN'



Step 21: Show the right angle at P



Step 22: Final completed hypocycloid construction

